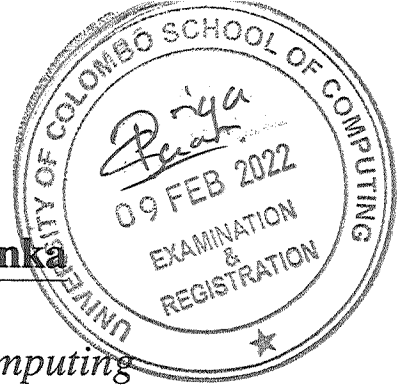




University of Colombo, Sri Lanka

University of Colombo School of Computing

Bachelor of Science in Computer Science

First Year Examination — Semester II - 2022

SCS1214 — Operating Systems I

(2 Hours)

Answer All Questions

Number of Pages = 11

Number of Questions = 4



To be completed by the candidate

Index Number

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Important Instructions

- The duration of the paper is 2 Hours.
- The medium of instructions and questions is English. Answer questions in English.
- This paper has 4 questions on 11 pages.
- Answer **all** questions.
- **Write your answers on and only on the space provided** on this question paper.
- Do not tear off any part of this answer book. Under no circumstances may this book (or any part of this book), used or unused, be removed from the Examination Hall by a candidate.
- Questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
- Any electronic device capable of storing and retrieving text, including electronic dictionaries and mobile phones, are **not allowed**.
- Non-programmable Calculators may be used.

To be completed by the examiners

1	
2	
3	
4	
Total	

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1. (a). The following code segment is written in the assembly instructions of the virtual machine used for the assignments of the SCS1214 course. There is a bug in **one** line of this code that causes the virtual machine to crash.

```
0 movv a 10
1 movv b 20
2 movv sp 18
3 call 18
4 out acc
5 add a b
6 out acc
7 halt

18 push a
19 push b
20 movv a 25
21 movv b 125
22 add a b
23 pop b
24 pop b
25 ret
```

- i. What is the content of the memory address 19 right after executing the instruction in line number 3?

[2 marks]

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- ii. Identify the line number with the bug?

[3 marks]

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- iii. Correct the buggy line of code and write down the corrected line in the space bellow.

[4 marks]

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- iv. Assume that the program executed correctly after you have changed the buggy line. What is the value of sp at the end of the program?

[4 marks]

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- v. Assume that the program executed correctly after you have changed the buggy line.
What is the output of the program?

[4 marks]

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- (b). The next state after a state transition of a process is given by the function $T(s, e)$, where s is the current state and e is an event.

A particular sequence of state transitions is given bellow.

$$T(T(T(X, e_1), e_2), e_3) = X$$

There are **no events** e_4 and e_5 such that,

$$T(T(X, e_4), e_5) = X$$

- i. What is e_1 ?

[2 marks]

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- ii. What is e_2 ?

[2 marks]

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- iii. What is e_3 ?

[2 marks]

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- iv. What is X ?

[2 marks]

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2. (a). Following program is compiled and executed on an x86 machine running Linux.

```
#include <stdio.h>

int main()
{
    int x,i;
    x=1;
    if(fork())
    {
        for(i=0; i < 1; i++)
        {
            x++;
            fork();
            printf("%d\n", x);
            x++;
        }
        fork();
    }
}
```

- i. What is the output of this program?

[5 marks]

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- ii. How many processes are created by this program?

[5 marks]

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- (b). A process running in ring 3 of an x86 machine executes the following instruction.

out ds, eax

What is the effect of executing this instruction?

[5 marks]

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- (c). A System has 14 instances of the resource type R and three processes, P_0, P_1, P_2 , that require R . The maximum requirements of R for each process and the current allocations at time t_0 are given in the following table.

	Maximum Need	Current Allocation
P_0	10	6
P_1	9	X
P_2	4	X

The system is in a safe state at t_0 .

- i. What is the largest possible value of X at t_0 ?

[5 marks]

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- ii. Give a safe sequences at t_0 when X has the largest possible value.

[5 marks]

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3. (a). What is meant by dispatch latency? List two (2) functionalities of a dispatcher in modern operating system.

Note Functionalities and characteristics are two different terms.

[5 marks]

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- (b). Suppose you are given a task to design your own CPU scheduling algorithm that should consist of the following features.

- Quickly respond to the requests that are submitted by process.
- Avoid keeping the CPU idle most of the time and get the CPU busy most of the time
- Minimize the time to execute a process.

Considering above features, what will be your targeting CPU scheduling criteria when developing your own CPU scheduler?

[5 marks]

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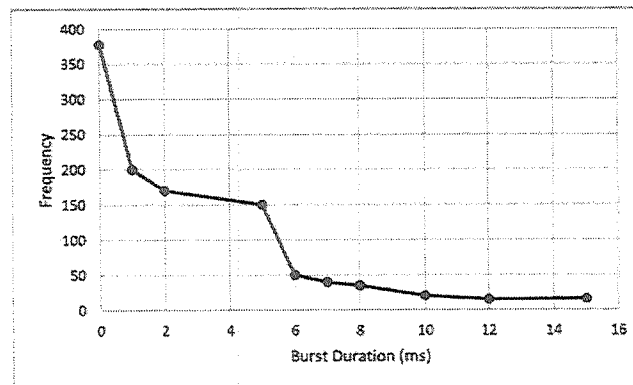
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- (c). What is the main drawback of priority scheduling? How can you mitigate such situation in scheduling?

[5 marks]

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- (d). The following histogram illustrates CPU-burst duration of a program.



By considering the above histogram, what is the nature of the process and what will be the best CPU scheduling algorithm that gives the lowest average waiting time for similar processes like above?

[5 marks]

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- (e). In rate monotonic CPU scheduling policy, priority is assigned based on its period. Which programs will get the highest and lowest priority?

[5 marks]

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4. (a). Argue the validity of the following statement.

Ideally, when we are given with more page frames, we would expect to have lesser page faults. For lower page frames we will get higher number of page faults. There is a linear relationship between page frames and page faults.

[5 marks]

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- (b). Assume that you are given a file system that consists of 32 bit addressing. Each data block in the file system are size of 1KB and pointer size is 4 byte. What is the **maximum** file size that can exist in your file system?

Assume that there are 10 direct pointers to data blocks, 3 indirect pointers, 2 double indirect pointers, and 1 triple indirect pointer defined in the file system.

[10 marks]

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- (c). i. Enhanced second change algorithm is improved by introducing reference bit and modify bit. Explain what these terms **reference bit** and **modify bit** refer to.

[2 marks]

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- ii. There exist two pages where in first page, only the reference bit is set. In the next page, modify bit is set. Which page will consume more time to replace when considering a page replacement candidate and why?

[3 marks]

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- (d). i. What is virtual memory?

[3 marks]

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ii. What are the two techniques that virtual memory is implemented?

[2 marks]

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